

Python and Git Class Assignments Setup Tutorial

Derek Harter

Department of Computer Science
East Texas A&M University

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Prerequisites

- A computer or laptop, not too old, with Windows 10/11, MacOS or Linux
 - It may be possible to use Android / iOS / ChromeBook, but not recommended nor supported
- Resource requirements are relatively modest to run Python interpreter
 - 4GB of memory, probably could work ok with less
 - Probably need only 5+GB of free disk space or less for Python and git / ssh tools. But assignments may need more beyond that.



Install a Python Interpreter

Derek Harter

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Install a Python Interpreter (Windows OS)

- Recommend downloading and installing official python.org package manager and distribution into Windows systems.
 - Anaconda Python Distribution is also a good choice, more features but more complex setup.
- Make sure using Python 3.
 - Should use Python 3.14 or newer
 - (as old as 3.11 or 3.12 might be acceptable).

python.org	https://www.python.org/downloads
Anaconda Python	https://www.anaconda.com/download



Install a Python Interpreter (MacOS)

- Recommend using Homebrew MacOS package manager to install Python 3 interpreter and git / ssh tools if needed.

```
$ xcode-select --install
$ /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/master/install.sh)"
$ brew install python
```

Homebrew <https://brew.sh>



Install a Python Interpreter (Linux OS)

- Probably the installed system default Python interpreter will work.
- Use system package manager (`apt`, `yum`) if not and need to install.
- On Linux systems, the interpreter executable may be named `python3` instead of simply `python`



Post Python Installation Checks and Tests

- On any system, you need to be able to run the Python 3 interpreter from a command line terminal, and it should be a relatively recent version of Python, e.g. 3.14.
- See links in written tutorial for how to open a command line terminal in Windows and MacOS if you are unfamiliar.
- First check if a python interpreter is on your path, and if so where it is located, using `where` or `which`

Windows systems

\$ `where python`

Linux or MacOS systems

\$ `which python`

- If available, check which version your default Python 3 is

\$ `python --version`
3.14.6



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Install 'git' and 'ssh' Tools

Derek Harter

Department of Computer Science
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Install git and ssh Tools (Windows OS)

- Windows OS does not come preinstalled with git nor ssh tools.
- Install from the official Git - Install for Windows download.
 - Override the default branch name and use 'main'
 - Ensure that "Use bundled OpenSSH" option is selected as you most likely do not have ssh tools on a Windows system yet.
 - Use "Checkout as-is, commit as-is" for line ending conversions.

Git - Install for Windows <https://git-scm.com/install/windows>



Install git and ssh tools (MacOS)

- You can install from the official Git - Install for MacOS.
- Or if you already have Homebrew prefer to use It

```
$ brew install git
```

Git - Install for MacOS <https://git-scm.com/install/mac>



Install git and ssh tools (Linux OS)

- git and ssh tools will most likely already be installed and available
- If not, try installing the build-essential package

```
$ sudo apt install build-essential
```



git and ssh Post Install Tests

- Test that you can access git and ssh from the command line.
- On Windows systems use the where command

```
C:\Users\etamustudent> where git
```

```
C:\Program Files\Git\cmd\git.exe
```

```
C:\Users\etamustudent> git --version
```

```
git version 2.55.0.windows.2
```

```
C:\Users\etamustudent>where ssh
```

```
C:\Windows\System32\OpenSSH\ssh.exe
```

```
C:\Users\etamustudent>ssh -V
```

```
OpenSSH_for_Windows_9.5p1, LibreSSL 3.8.2
```



git and ssh Post Install Tests

- On Linux and MacOs, use `which`, and remember the interpreter may be named `python3` instead of `python`

```
$ which git  
/usr/bin/git
```

```
$ git --version  
git version 2.53.0
```

```
$ which ssh  
/usr/bin/ssh
```

```
$ ssh -V  
OpenSSH_10.2p1 Ubuntu-2ubuntu3.2, OpenSSL 3.5.5 27 Jan 2026
```



git Global Configuration

- On all systems you should set up basic git global configurations
- See Customizing Git - Git Configuration for a more detailed discussion.

```
C:\Users\etamustudent> git config --global user.name "ETAMU Student"
```

```
C:\Users\etamustudent> git config --global user.email "estudent@leomail.etamu.edu"
```

```
C:\Users\etamustudent> git config --list
```

```
...
```

```
user.name=ETAMU Student
```

```
user.email=estudent@leomail.etamu.edu
```

- You can list your global git configuration using the `--list` flag.
- Use it to check that you correctly set your `user.name` and `user.email`



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Create GitHub Account and Set Up ssh Keypair

- Create a GitHub account for yourself GitHub.com
 - Use your Leo Mail e-mail address as primary e-mail
 - Ensure that your git config user.email matches your GitHub primary email address.
- For all OS, generate a default ssh public / private Keypair

```
C:\Users\etamustudent> ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (C:\Users\etamustudent/.ssh/id_ed25519):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in C:\Users\etamustudent/.ssh/id_ed25519
Your public key has been saved in C:\Users\etamustudent/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:LeDf085S+F4HGAhzD73pikAAVH1ZYgUG1bUYWdiqI4s etamustudent@windemo
The key's randomart image is:
+--[ED25519 256]--+
|oo.o+B#o.      |
| ..O*+=.+      |
| ...o.. oo      |
| .o . .oo      |
| .. . Soo .     |
| . o. .oo .     |
|. o .. ..+. .   |
|E. . o.oo .     |
|      +=.       |
+-----[SHA256]-----+
```



Create GitHub Account and Set Up ssh Keypair

- We generated key into C:\Users\etamustudent\.ssh\id_ed25519 file
 - Public key is in id_ed25519.pub
 - Don't use a passphrase (just hit return)
 - Copy the public key, and create a **New SSH Key** on GitHub and paste your key.
- IMPORTANT** Confirm you can authenticate with your key to GitHub

```
C:\Users\etamustudent> ssh git@github.com
The authenticity of host 'github.com (140.82.113.4)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TUJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
PTY allocation request failed on channel 0
Hi etamustudent! You've successfully authenticated, but GitHub does not provide shell access.
Connection to github.com closed.
```



Text Editors / IDE for Python Programming Assignments

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Text Editors / IDE for Python Programming Assignments

- You will need a programming text editor to edit and create assignments.
 - Popular choices include
 - Notepad++
 - Sublime Text
 - TextMate
- We recommended installing VSCode if you don't have a preference
 - A full blown extendable IDE
 - Built in gui for git revision control management
 - Install the "Python" extension pack to get main functions needed.

Visual Studio Code <https://code.visualstudio.com/download>

